

Mathematical Foundations of the WCS-GNCCP Molecular Graph Kernel with Hierarchical 2D–3D Alignment

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Abstract—We present the mathematical foundations of a hierarchical molecular similarity kernel that combines 2D topological matching with 3D geometric alignment. The 2D stage solves a Weighted Common Subgraph (WCS) problem via Graded Non-Convexity and Concavity Procedure (GNCCP) with Frank-Wolfe optimization, producing both a similarity score and a bijective atom correspondence. We derive a closed-form structural gradient (Theorem 1), prove a rank-one factorization (Lemma 1), and establish scale invariance via Frobenius normalization (Proposition 4). The bijective map produced by Stage 1 reduces the 3D alignment (Stage 2) to a single $O(n)$ Kabsch rotation, yielding a hybrid kernel with per-pair complexity dominated by the 2D stage at $O(T_\zeta T_{FW} n^3)$ — empirically competitive with Random Walk and Optimal Assignment kernels on drug-like molecules.

I. PROBLEM FORMULATION

Let G and H be two molecular graphs with M and N atoms, respectively. Their topology is encoded by weighted adjacency matrices $\mathbf{A}_G \in \mathbb{R}^{M \times M}$ and $\mathbf{A}_H \in \mathbb{R}^{N \times N}$, where each entry reflects bond type (single, double, triple, aromatic) and topology (ring membership, conjugation). A cost matrix $\mathbf{C} \in \mathbb{R}^{M \times N}$ encodes pairwise atom-feature dissimilarity.

Definition 1 (Partial Permutation Polytope). *For an integer $L \leq \min(M, N)$, the partial permutation polytope is*

$$\mathcal{P}_L = \{ \mathbf{X} \in \{0, 1\}^{M \times N} \mid \mathbf{X} \mathbf{1}_N \leq \mathbf{1}_M, \mathbf{X}^T \mathbf{1}_M \leq \mathbf{1}_N, \mathbf{1}_M^T \mathbf{X} \mathbf{1}_N = L \}, \quad (1)$$

and its convex relaxation $\mathcal{D}_L$ replaces $\{0, 1\}^{M \times N}$ with $[0, 1]^{M \times N}$. An entry $X_{ij} = 1$ indicates that atom i in G is matched to atom j in H .

Definition 2 (WCS Objective). *The Weighted Common Subgraph (WCS) objective is*

$$F(\mathbf{X}) = \alpha f(\mathbf{X}) + (1 - \alpha) \text{tr}(\mathbf{C}^T \mathbf{X}), \quad (2)$$

where $\alpha \in [0, 1]$ balances structure against atom features,

$$f(\mathbf{X}) = \|\mathbf{U} \circ \mathbf{A}_G - \mathbf{X} \mathbf{A}_H \mathbf{X}^T\|_F^2 \quad (3)$$

is the structural mismatch term, $\circ$ denotes the Hadamard product, and

$$\mathbf{U} = \mathbf{X} \mathbf{1}_{N \times N} \mathbf{X}^T \in \mathbb{R}^{M \times M} \quad (4)$$

is the co-matching indicator.

II. COMPUTATIONAL IDENTITY FOR $\mathbf{U}$

The matrix $\mathbf{U}$ in (4) involves the $N \times N$ all-ones matrix, requiring $O(MN^2 + M^2N)$ work. We show that $\mathbf{U}$ admits a rank-one factorization that reduces this to $O(MN + M^2)$.

Lemma 1 (Rank-One Factorization of $\mathbf{U}$). *Let $\mathbf{r} = \mathbf{X} \mathbf{1}_N \in \mathbb{R}^M$ be the row-sum vector. Then*

$$\mathbf{U} = \mathbf{X} \mathbf{1}_{N \times N} \mathbf{X}^T = \mathbf{r} \mathbf{r}^T. \quad (5)$$

Proof. Factor the all-ones matrix as $\mathbf{1}_{N \times N} = \mathbf{1}_N \mathbf{1}_N^T$:

$$\begin{aligned} \mathbf{U} &= \mathbf{X} (\mathbf{1}_N \mathbf{1}_N^T) \mathbf{X}^T \\ &= (\mathbf{X} \mathbf{1}_N) (\mathbf{X} \mathbf{1}_N)^T = \mathbf{r} \mathbf{r}^T. \end{aligned} \quad (6)$$

□

Remark 1. $\mathbf{U}$ is symmetric PSD with rank one. For binary $\mathbf{X} \in \mathcal{P}_L$ each $r_i \in \{0, 1\}$, so $U_{ik} = 1$ iff both atoms i, k are matched.

III. GRADIENT OF THE STRUCTURAL TERM

We derive $\nabla_{\mathbf{X}} f$ for the structural cost (3).

Theorem 1 (Structural Gradient). *Define the residual $\mathbf{R} = \mathbf{U} \circ \mathbf{A}_G - \mathbf{X} \mathbf{A}_H \mathbf{X}^T$. Then*

$$\nabla_{\mathbf{X}} f = 4(\mathbf{R} \circ \mathbf{A}_G) \mathbf{X} \mathbf{1}_{N \times N} - 4 \mathbf{R} \mathbf{X} \mathbf{A}_H. \quad (7)$$

Moreover, using Lemma 1:

$$(\mathbf{R} \circ \mathbf{A}_G) \mathbf{X} \mathbf{1}_{N \times N} = \left[(\mathbf{R} \circ \mathbf{A}_G) \mathbf{X} \right] \mathbf{1}_N \mathbf{1}_N^T, \quad (8)$$

a rank-one matrix whose i -th row repeats $\sum_j [(\mathbf{R} \circ \mathbf{A}_G) \mathbf{X}]_{ij}$.

Proof. Write $f = \text{tr}(\mathbf{R}^T \mathbf{R})$. By the chain rule,

$$df = 2 \text{tr}(\mathbf{R}^T d\mathbf{R}). \quad (9)$$

Part 1 (Hadamard term). Since $\mathbf{A}_G$ is constant,

$$d(\mathbf{U} \circ \mathbf{A}_G) = (d\mathbf{U}) \circ \mathbf{A}_G. \quad (10)$$

From $\mathbf{U} = \mathbf{r} \mathbf{r}^T$ with $\mathbf{r} = \mathbf{X} \mathbf{1}_N$:

$$d\mathbf{U} = (d\mathbf{X} \mathbf{1}_N) \mathbf{r}^T + \mathbf{r} (d\mathbf{X} \mathbf{1}_N)^T. \quad (11)$$

Part 2 (Bilinear term). By the product rule,

$$\begin{aligned} d(\mathbf{X}\mathbf{A}_H\mathbf{X}^T) &= (d\mathbf{X})\mathbf{A}_H\mathbf{X}^T \\ &\quad + \mathbf{X}\mathbf{A}_H(d\mathbf{X})^T. \end{aligned} \quad (12)$$

Part 3 (Combining). Substitute (10)–(12) into (9) and apply the Hadamard trace identity

$$\text{tr}(\mathbf{A}^T(\mathbf{B} \circ \mathbf{C})) = \text{tr}((\mathbf{A} \circ \mathbf{B})^T \mathbf{C}).$$

Collecting terms in $d\mathbf{X}$: the Hadamard contribution yields $(\mathbf{R} \circ \mathbf{A}_G)\mathbf{X}\mathbf{1}_{N \times N}$ and the bilinear contribution yields $-\mathbf{R}\mathbf{X}\mathbf{A}_H$. Since $\mathbf{R}$, $\mathbf{A}_G$, $\mathbf{U}$, and $\mathbf{X}\mathbf{A}_H\mathbf{X}^T$ are all symmetric, each of the two terms in (11) and (12) contributes equally, producing the factor of 4. $\square$

Remark 2. A common error is to differentiate $\mathbf{U} \circ \mathbf{A}_G$ as if it were the matrix product $\mathbf{U}\mathbf{A}_G$. The Hadamard derivative is $d(\mathbf{U} \circ \mathbf{A}_G) = (d\mathbf{U}) \circ \mathbf{A}_G$ (element-wise), not $(d\mathbf{U})\mathbf{A}_G$.

IV. THE GNCCP CONTINUATION METHOD

Direct minimization of F over $\mathcal{D}_L$ is nonconvex. GNCCP introduces $\zeta \in [-1, 1]$ to deform the problem from convex to concave.

Definition 3 (GNCCP Objective). For $\zeta \in [-1, 1]$, define

$$J_\zeta(\mathbf{X}) = \begin{cases} (1 - \zeta) F(\mathbf{X}) + \zeta \|\mathbf{X}\|_F^2 & \zeta \geq 0, \\ (1 + \zeta) F(\mathbf{X}) + \zeta \|\mathbf{X}\|_F^2 & \zeta < 0. \end{cases} \quad (13)$$

Proposition 1 (Endpoint Properties).

- 1) $\zeta = +1$: $J_{+1} = \|\mathbf{X}\|_F^2$ is strictly convex. Its unique minimizer over $\mathcal{D}_L$ is the uniform matrix $\mathbf{X}^{(0)} = \frac{L}{MN} \mathbf{1}_{M \times N}$.
- 2) $\zeta = 0$: $J_0 = F(\mathbf{X})$, the original WCS objective.
- 3) $\zeta = -1$: $J_{-1} = -\|\mathbf{X}\|_F^2$ is strictly concave. Its minimizers over $\mathcal{D}_L$ are vertices, i.e., partial permutation matrices in $\mathcal{P}_L$.

Proof. (1) $\|\mathbf{X}\|_F^2 = \sum_{ij} X_{ij}^2$ is strictly convex. The unique minimizer distributes mass uniformly: $X_{ij} = L/(MN)$ satisfies all constraints and minimizes the sum of squares by Schur convexity.

(2) Direct substitution.

(3) $-\|\mathbf{X}\|_F^2$ is strictly concave. A strictly concave function attains its minimum over a compact convex polytope at a vertex. The vertices of $\mathcal{D}_L$ are exactly the elements of $\mathcal{P}_L$. $\square$

Proposition 2 (Gradient of J_ζ).

$$\nabla_{\mathbf{X}} J_\zeta = \begin{cases} (1 - \zeta) \nabla_{\mathbf{X}} F + 2\zeta \mathbf{X} & \zeta \geq 0, \\ (1 + \zeta) \nabla_{\mathbf{X}} F + 2\zeta \mathbf{X} & \zeta < 0, \end{cases} \quad (14)$$

where $\nabla_{\mathbf{X}} F = \alpha \nabla_{\mathbf{X}} f + (1 - \alpha) \mathbf{C}$ and $\nabla_{\mathbf{X}} f$ is given by Theorem 1.

Proof. $\nabla \|\mathbf{X}\|_F^2 = 2\mathbf{X}$. Apply linearity to (13). $\square$

V. FRANK-WOLFE OPTIMIZATION

For each fixed ζ we solve $\min_{\mathbf{X} \in \mathcal{D}_L} J_\zeta(\mathbf{X})$ via Frank-Wolfe (conditional gradient).

Definition 4 (Linear Minimization Oracle). At iterate $\mathbf{X}_t$, the FW direction is

$$\mathbf{Y}_t = \arg \min_{\mathbf{Y} \in \mathcal{D}_L} \langle \nabla J_\zeta(\mathbf{X}_t), \mathbf{Y} \rangle. \quad (15)$$

Since the minimum of a linear function over a polytope is attained at a vertex, $\mathbf{Y}_t \in \mathcal{P}_L$. The optimal vertex is found by the Hungarian algorithm in $O(\min(M, N)^3)$ time.

Theorem 2 (Frank-Wolfe Duality Gap). Let J_ζ be convex on $\mathcal{D}_L$ and let $\mathbf{X}^* = \arg \min_{\mathcal{D}_L} J_\zeta$. The Frank-Wolfe gap

$$g_t = \langle \nabla J_\zeta(\mathbf{X}_t), \mathbf{X}_t - \mathbf{Y}_t \rangle \quad (16)$$

satisfies: (i) $g_t \geq 0$; (ii) $g_t \geq J_\zeta(\mathbf{X}_t) - J_\zeta(\mathbf{X}^*)$; (iii) $g_t = 0 \Leftrightarrow \mathbf{X}_t$ is optimal.

Proof. (i) $\mathbf{Y}_t$ minimizes the linear form over $\mathcal{D}_L$ and $\mathbf{X}_t \in \mathcal{D}_L$, so $\langle \nabla J_\zeta, \mathbf{Y}_t \rangle \leq \langle \nabla J_\zeta, \mathbf{X}_t \rangle$, giving $g_t \geq 0$.

(ii) By convexity:

$$J_\zeta(\mathbf{X}^*) \geq J_\zeta(\mathbf{X}_t) + \langle \nabla J_\zeta(\mathbf{X}_t), \mathbf{X}^* - \mathbf{X}_t \rangle. \quad (17)$$

Since $\mathbf{Y}_t$ is the minimizer and $\mathbf{X}^* \in \mathcal{D}_L$:

$$\langle \nabla J_\zeta(\mathbf{X}_t), \mathbf{X}^* - \mathbf{X}_t \rangle \geq \langle \nabla J_\zeta(\mathbf{X}_t), \mathbf{Y}_t - \mathbf{X}_t \rangle = -g_t. \quad (18)$$

Combining (17)–(18): $g_t \geq J_\zeta(\mathbf{X}_t) - J_\zeta(\mathbf{X}^*)$.

(iii) If $\mathbf{X}_t$ is optimal, first-order optimality gives $\langle \nabla J_\zeta, \mathbf{Y} - \mathbf{X}_t \rangle \geq 0 \ \forall \mathbf{Y} \in \mathcal{D}_L$, so $g_t = 0$. Conversely, $g_t = 0$ with (ii) gives $J_\zeta(\mathbf{X}_t) \leq J_\zeta(\mathbf{X}^*)$. $\square$

Remark 3. When J_ζ is nonconvex ($\zeta < 0$), the gap (16) is no longer a certified suboptimality bound, but $g_t = 0$ still implies $\mathbf{X}_t$ is a stationary point. We use $g_t < \varepsilon$ as early termination for all ζ .

VI. KERNEL PROPERTIES

Proposition 3 (Exponential Graph Similarity Kernel). The exponential mapping

$$k(G, H) = \exp\left(-\frac{\gamma F_{GH}^*}{L}\right) \quad (19)$$

where $\gamma > 0$ is a bandwidth parameter and F_{GH}^* is the optimal WCS objective, defines a discriminative similarity measure satisfying:

- 1) $k(G, H) \in (0, 1]$ for all graph pairs;
- 2) $k(G, G) = 1$ when G is matched to itself;
- 3) k is monotonically decreasing in F_{GH}^* .

Remark 4. The kernel (19) is generally not positive semi-definite in the Mercer sense, since optimal-assignment and graph-edit distances do not satisfy conditional negative definiteness (cf. Schoenberg [1]). This is not a limitation for distance-based classifiers such as k -NN (which rely on ranking, not inner products) or for SVM with eigenvalue ridge correction ($\mathbf{K}_{psd} = \mathbf{K} + \mu \mathbf{I}$, $\mu \geq |\lambda_{\min}|$).

Remark 5. γ controls bandwidth: small γ spreads kernel values (all pairs appear similar), large γ concentrates mass on near-identical pairs. We default to $\gamma = 0.5$ and normalize by $\max(L, 1)$ for size invariance.

VII. SCALE INVARIANCE

Proposition 4 (Frobenius Normalization). *Let $\hat{\mathbf{A}}_G = \mathbf{A}_G / \|\mathbf{A}_G\|_F$ and $\hat{\mathbf{A}}_H = \mathbf{A}_H / \|\mathbf{A}_H\|_F$. Then*

$$\hat{f}(\mathbf{X}) = \|\mathbf{U} \circ \hat{\mathbf{A}}_G - \mathbf{X} \hat{\mathbf{A}}_H \mathbf{X}^T\|_F^2 \leq 2 \quad (20)$$

for all $\mathbf{X} \in \mathcal{D}_L$, independently of the number of bonds.

Proof. By the identity $\|\mathbf{A} - \mathbf{B}\|_F^2 = \|\mathbf{A}\|_F^2 + \|\mathbf{B}\|_F^2 - 2\langle \mathbf{A}, \mathbf{B} \rangle$, it suffices to bound each squared norm.

For the Hadamard term, using $\|\mathbf{A} \circ \mathbf{B}\|_F \leq \|\mathbf{A}\|_{\max} \|\mathbf{B}\|_F$ with $U_{ik} = r_i r_k \leq 1$ for $\mathbf{X} \in \mathcal{D}_L$:

$$\|\mathbf{U} \circ \hat{\mathbf{A}}_G\|_F \leq 1 \cdot \|\hat{\mathbf{A}}_G\|_F = 1. \quad (21)$$

For the bilinear term, since $\|\mathbf{X}\|_2 \leq 1$ for sub-stochastic $\mathbf{X}$:

$$\|\mathbf{X} \hat{\mathbf{A}}_H \mathbf{X}^T\|_F \leq \|\mathbf{X}\|_2^2 \|\hat{\mathbf{A}}_H\|_F \leq 1. \quad (22)$$

Therefore $\hat{f} \leq \|\mathbf{U} \circ \hat{\mathbf{A}}_G\|_F^2 + \|\mathbf{X} \hat{\mathbf{A}}_H \mathbf{X}^T\|_F^2 \leq 2$. $\square$

Remark 6. Without normalization, $f(\mathbf{X})$ scales with $\|\mathbf{A}_G\|_F^2 + \|\mathbf{A}_H\|_F^2$, which grows linearly in the number of bonds. For 30 bonds at weight ~ 1.5 , $\|\mathbf{A}_G\|_F \approx 9.5$, making the structural cost $\sim 90\times$ larger than the appearance cost $\text{tr}(\mathbf{C}^T \mathbf{X}) \in [0, L]$. Normalization restores the balance intended by α .

VIII. PROJECTION ONTO $\mathcal{D}_L$

Proposition 5 (Sinkhorn-Knopp Projection). *Let $\mathbf{X} \in \mathbb{R}_{\geq 0}^{M \times N}$ with $\|\mathbf{X}\|_1 > 0$. The iteration*

1. **Scale:** $\mathbf{X} \leftarrow \mathbf{X} \cdot (L / \|\mathbf{X}\|_1)$.
2. **Row cap:** if $(\mathbf{X} \mathbf{1}_N)_i > 1$, divide row i by $(\mathbf{X} \mathbf{1}_N)_i$.
3. **Col cap:** if $(\mathbf{X}^T \mathbf{1}_M)_j > 1$, divide col j by $(\mathbf{X}^T \mathbf{1}_M)_j$.
4. Repeat 1–3 until convergence.

converges to a feasible point in $\mathcal{D}_L$ when $L \leq \min(M, N)$.

Proof sketch. Steps 2–3 are alternating projections onto $\mathcal{R} = \{\mathbf{X} \geq 0 : \mathbf{X} \mathbf{1} \leq \mathbf{1}\}$ and $\mathcal{C} = \{\mathbf{X} \geq 0 : \mathbf{X}^T \mathbf{1} \leq \mathbf{1}\}$, both convex and closed. Step 1 enforces $\|\mathbf{X}\|_1 = L$. By Dykstra’s theorem, this converges when the intersection is non-empty. The uniform matrix $\frac{L}{MN} \mathbf{1}_{M \times N}$ lies in all three constraint sets, so the intersection is non-empty. $\square$

IX. COMPUTATIONAL COMPLEXITY ANALYSIS

We provide an honest, operation-level complexity analysis of each Frank-Wolfe iteration.

Theorem 3 (Per-Iteration Complexity). *Let M, N denote the atom counts and let d be the maximum vertex degree of the molecular graphs. One Frank-Wolfe iteration of Algorithm 1 has complexity*

$$O\left(\underbrace{M^2 N}_{\mathbf{X} \mathbf{A}_H \mathbf{X}^T} + \underbrace{\min(M, N)^3}_{\text{Hungarian}}\right). \quad (23)$$

For $M \approx N$, this simplifies to $\Theta(n^3)$.

Proof. We enumerate the dominant operations per iteration:

(a) *Row sums:* $\mathbf{r} = \mathbf{X} \mathbf{1}_N$ costs $O(MN)$.

(b) *Co-matching indicator (Lemma 1):* $\mathbf{U} = \mathbf{r} \mathbf{r}^T$ costs $O(M^2)$. Without Lemma 1, the naïve computation via (4) costs $O(MN^2 + M^2 N)$.

(c) *Bilinear form:* $\mathbf{P} = \mathbf{X} \mathbf{A}_H$ costs $O(M \cdot \text{nnz}(\mathbf{A}_H))$; for molecular graphs with bounded degree d , $\text{nnz}(\mathbf{A}_H) = O(dN)$, so $\mathbf{P}$ costs $O(MdN) \approx O(MN)$. However, $\mathbf{P} \mathbf{X}^T$ is an $(M \times N)(N \times M)$ product costing $O(M^2 N)$. This is irreducible for dense $\mathbf{X}$.

(d) *Gradient terms:* $(\mathbf{R} \circ \mathbf{A}_G) \mathbf{X}$ costs $O(\text{nnz}(\mathbf{A}_G) \cdot N) = O(dMN)$; $\mathbf{R} \mathbf{X} \mathbf{A}_H$ costs $O(M^2 N + MdN)$.

(e) *Hungarian algorithm:* $O(\min(M, N)^3)$; this is tight (Kuhn–Munkres).

(f) *Projection:* $O(MN)$ per Sinkhorn iteration, bounded at 20 iterations.

The dominant terms are (c) and (e), giving (23). $\square$

Corollary 1 (Reduction from Lemma 1). *The rank-one factorization eliminates one $O(MN^2 + M^2 N)$ operation per gradient evaluation, reducing it to $O(MN + M^2)$. For $M = N = n$, this is a reduction from $O(n^3)$ to $O(n^2)$ for the $\mathbf{U}$ computation alone. The overall per-iteration complexity remains $\Theta(n^3)$ due to the bilinear form (c) and Hungarian assignment (e).*

Corollary 2 (Total Per-Pair Complexity). *With T_ζ GNCCP outer iterations and T_{FW} Frank-Wolfe inner iterations, the total per-pair cost is*

$$O(T_\zeta T_{\text{FW}} n^3). \quad (24)$$

In practice, early termination via the FW duality gap (Theorem 2) reduces the effective iteration count well below the worst case.

Remark 7 (Practical Competitiveness). *For drug-like molecules ($n \leq 50$), the cubic cost per iteration is sub-millisecond. With $(T_\zeta, T_{\text{FW}}) = (4, 4)$ and hydrogen removal ($n \approx 30$ heavy atoms), the measured per-pair time is ~ 6 ms on CPU — competitive with Random Walk ($O(n^3)$, ~ 10 – 30 ms) and Optimal Assignment kernels ($O(n^3)$, ~ 10 – 50 ms), while producing a richer output (bijective atom map in addition to similarity score).*

X. STAGE 2: 3D ALIGNMENT VIA KABSCH ROTATION

The bijective atom correspondence $\mathbf{X}^*$ from Stage 1 induces a set of L matched atom pairs $\mathcal{M} = \{(i, j) : X_{ij}^* = 1\}$. We exploit this mapping to perform 3D alignment in $O(L)$ time.

Definition 5 (Matched Coordinate Sets). *Let $\mathbf{p}_1, \dots, \mathbf{p}_L \in \mathbb{R}^3$ and $\mathbf{q}_1, \dots, \mathbf{q}_L \in \mathbb{R}^3$ be the 3D coordinates of the matched atoms in molecules G and H respectively, ordered according to $\mathcal{M}$. Define centered coordinates $\bar{\mathbf{p}}_l = \mathbf{p}_l - \boldsymbol{\mu}_P$ and $\bar{\mathbf{q}}_l = \mathbf{q}_l - \boldsymbol{\mu}_Q$, where $\boldsymbol{\mu}_P = \frac{1}{L} \sum_l \mathbf{p}_l$ and $\boldsymbol{\mu}_Q = \frac{1}{L} \sum_l \mathbf{q}_l$.*

Theorem 4 (Kabsch Alignment). *The rotation matrix $\mathbf{R}^* \in \text{SO}(3)$ that minimizes the RMSD*

$$\text{RMSD}^2 = \frac{1}{L} \sum_{l=1}^L \|\mathbf{R}\bar{\mathbf{p}}_l - \bar{\mathbf{q}}_l\|^2 \quad (25)$$

is given by

$$\mathbf{R}^* = \mathbf{V} \begin{pmatrix} 1 & & \\ & 1 & \\ & & \det(\mathbf{V}\mathbf{V}^T) \end{pmatrix} \mathbf{W}^T, \quad (26)$$

where $\mathbf{H} = \bar{\mathbf{P}}^T \bar{\mathbf{Q}} \in \mathbb{R}^{3 \times 3}$ is the cross-covariance matrix and $\mathbf{H} = \mathbf{W}\mathbf{\Sigma}\mathbf{V}^T$ is its SVD.

Proof. Standard; see Kabsch (1976). The key observation is that $\mathbf{H}$ is always 3×3 regardless of L , so its SVD costs $O(1)$. The total cost is dominated by forming $\mathbf{H}$, which requires $O(L)$ vector operations. $\square$

Corollary 3 (Stage 2 Complexity). *Given the atom map $\mathcal{M}$ from Stage 1, the 3D alignment and RMSD computation cost $O(L) = O(n)$.*

Definition 6 (Hybrid Kernel). *The two-stage similarity measure combines 2D topological and 3D geometric information:*

$$k_{\text{hybrid}}(G, H) = (1 - \beta) k_{2D}(G, H) + \beta k_{3D}(G, H), \quad (27)$$

where $k_{2D} = \exp(-\gamma_{2D} F^*/L)$ is the topological kernel from (19), $k_{3D} = \exp(-\gamma_{3D} \cdot \text{RMSD})$ is the geometric kernel, and $\beta \in [0, 1]$ is cross-validated per target.

Remark 8 (Overall Pipeline Complexity). *The total per-pair cost of the hierarchical pipeline is*

$$\underbrace{O(T_\zeta T_{\text{FW}} n^3)}_{\text{Stage 1: 2D kernel}} + \underbrace{O(n)}_{\text{Stage 2: 3D alignment}} = O(T_\zeta T_{\text{FW}} n^3), \quad (28)$$

completely dominated by Stage 1. The 2D stage serves as a search-space reduction gate: only candidates with $k_{2D} > \tau$ (a tunable threshold) proceed to Stage 2. For a library of P molecules with filtration rate ϕ , the total screening cost is

$$\underbrace{O\left(\binom{P}{2} T_\zeta T_{\text{FW}} n^3\right)}_{\text{2D pairwise kernel matrix}} + \underbrace{O(\phi P^2 n)}_{\text{3D alignment of filtered pairs}}. \quad (29)$$

With aggressive filtering ($\phi \ll 1$), the 3D cost becomes negligible.

XI. RELABELING INVARIANCE

Proposition 6 (Relabeling Invariance). *Let $\mathbf{P} \in \{0, 1\}^{M \times M}$ and $\mathbf{Q} \in \{0, 1\}^{N \times N}$ be permutation matrices, and let $\mathbf{A}'_G = \mathbf{P}\mathbf{A}_G\mathbf{P}^T$, $\mathbf{A}'_H = \mathbf{Q}\mathbf{A}_H\mathbf{Q}^T$. Then for any $\mathbf{X} \in \mathcal{D}_L$,*

$$F(\mathbf{P}\mathbf{X}\mathbf{Q}^T; \mathbf{A}'_G, \mathbf{A}'_H, \mathbf{C}') = F(\mathbf{X}; \mathbf{A}_G, \mathbf{A}_H, \mathbf{C}), \quad (30)$$

where $\mathbf{C}' = \mathbf{P}\mathbf{C}\mathbf{Q}^T$.

Proof. Let $\mathbf{X}' = \mathbf{P}\mathbf{X}\mathbf{Q}^T$. Then $\mathbf{X}'\mathbf{A}'_H(\mathbf{X}')^T = \mathbf{P}\mathbf{X}\mathbf{A}_H\mathbf{X}^T\mathbf{P}^T$ and $\mathbf{U}' \circ \mathbf{A}'_G = \mathbf{P}(\mathbf{U} \circ \mathbf{A}_G)\mathbf{P}^T$. The Frobenius norm is unitarily invariant: $\|\mathbf{P}\mathbf{M}\mathbf{P}^T\|_F = \|\mathbf{M}\|_F$. The trace term satisfies $\text{tr}((\mathbf{C}')^T \mathbf{X}') = \text{tr}(\mathbf{C}^T \mathbf{X})$ by the cyclic property of trace. $\square$

XII. FULL ALGORITHM

Algorithm 1 WCS-GNCCP with Frank-Wolfe

Require: $\mathbf{A}_G, \mathbf{A}_H$; cost $\mathbf{C}$; size L ; weights α, γ

Ensure: Similarity score $k \in (0, 1]$ and atom map $\mathcal{M}$

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1:  $\hat{\mathbf{A}}_G \leftarrow \mathbf{A}_G / \|\mathbf{A}_G\|_F$ ;  $\hat{\mathbf{A}}_H \leftarrow \mathbf{A}_H / \|\mathbf{A}_H\|_F$  {Prop. 4}
2:  $\hat{\mathbf{C}} \leftarrow (\mathbf{C} - C_{\min}) / (C_{\max} - C_{\min})$ 
3:  $\mathbf{X} \leftarrow \frac{L}{MN} \mathbf{1}_{M \times N}$  {Prop. 1(1)}
4: for  $\zeta = 1, 1 - \Delta\zeta, \dots, -1$  do
5:   for  $t = 1, \dots, T_{\text{FW}}$  do
6:      $\nabla \leftarrow \nabla_{\mathbf{X}} J_\zeta$  {Prop. 2}
7:      $\mathbf{Y} \leftarrow \text{Hungarian}(\nabla)$  {Def. 4}
8:      $g \leftarrow \langle \nabla, \mathbf{X} - \mathbf{Y} \rangle$  {Thm. 2}
9:     if  $g < \varepsilon$  then
10:      break
11:   end if
12:    $\lambda \leftarrow \text{ArmijoLS}(\mathbf{X}, \mathbf{Y}, J_\zeta)$ 
13:    $\mathbf{X} \leftarrow \text{Sinkhorn}(\mathbf{X} + \lambda(\mathbf{Y} - \mathbf{X}), L)$  {Prop. 5}
14: end for
15: end for
16:  $\mathbf{X}^* \leftarrow \text{Hungarian}(-\mathbf{X})$  {discretize}
17:  $\mathcal{M} \leftarrow \{(i, j) : X^*_{ij} = 1\}$  {atom map}
18:  $F^* \leftarrow F(\mathbf{X}^*)$  {Def. 2}
19: return  $k = \exp(-\gamma F^*/L)$ ,  $\mathcal{M}$  {Prop. 3}
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